

Sarah Chen

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Education

Georgia Institute of Technology

Atlanta, GA

B.S. IN COMPUTER SCIENCE | GPA: 3.75/4.00

Aug. 2017 - May 2021

- Current Coursework: Introduction to Graph Theory, Stochastic Processes, Deep Learning, Privacy Technology, Policy, and Law
- Theory Coursework: Advanced Algorithms for Machine Learning, Algorithms Honors, Second Course in Linear Algebra
- Intelligence Coursework: Artificial Intelligence, Machine Learning

Experience

Goldman Sachs

New York, NY

SECURITIES STRATEGIST SUMMER ANALYST

May 2019 - Aug. 2019

- Equities Derivatives Automation Desk Strategist
 - Integrated trader-to-trader workflow in the desk's automatic pricing platform, cutting quoting time from 25 minutes to under a second.
 - Enhanced the platform coverage to include pricing for many options strategies, and added percentage strike and spot reference features.
- FICCS Commodities Desk Strategist
 - Implemented memoization for portfolio Greek valuations, cutting time used to compare risks from different hedging strategies by 500%.
 - Built sensors that tracked changes in internal risk and EIA natural gas and petroleum storage, improving our real-time trading strategies.
 - Performed correlation analysis using moment matching on power and natural gas futures contracts, used for more precise trade pricing.

Institute for Data Engineering and Science

Atlanta, GA

FORMAL FOUNDATIONS OF ALGORITHMIC MATTER AND EMERGENT COMPUTATION RESEARCH ASSISTANT

Aug. 2019 - Present

- Simulate self-organizing particle systems by implementing various distributed, randomized algorithms in which nearest neighbors interact and the ensemble is trying to achieve some collective goal. Sample tasks to be explored include: alignment, separation and locomotion.

Art of Problem Solving

San Diego, CA

TEACHING ASSISTANT

Aug. 2019 - Present

- Grade homework solutions for several intermediate and advanced mathematics courses, such as algebra, counting and probability, number theory, and geometry, as well as a Python programming course.

Georgia Tech College of Computing

Atlanta, GA

DISCRETE MATH HONORS HEAD TEACHING ASSISTANT

Jan. 2018 - May 2019

- Assisted professor in preparing lesson plans, teaching review sessions, writing homework and exam solutions, holding biweekly office hours, and grading homework and exams.

Projects

Empowoment

Chapel Hill, NC

PEARL HACKS

Feb. 2019

- Used Dialogflow to train an AI Chatbot that encourages women to use fewer unnecessarily apologetic or uncertain phrases, such as "maybe."
- Integrated bot with Facebook Messenger so that it could be used in group chats to better support our client base.
- Best Use of Google Cloud Platform, Best Domain Registered with Domain.com, and Best Use of Splunk awards.

SafePath

Atlanta, GA

HACKGT V

Oct. 2018

- Used Android Studio to create a mobile app that focuses on personal safety with route tracking: if users step outside a predetermined "safe zone," an alert is sent to his or her emergency contact via text.
- Integrated app with Firebase, Google Awareness API, and Esri ArcGIS API.

Homeless Shelter Locator

Atlanta, GA

OBJECTS AND DESIGN COURSE

Jan. 2018 - May 2018

- Used Android Studio to create a mobile app through which users can find nearby homeless shelters that meet their needs.
- Worked on Firebase integration, account lockout, push notifications, password recovery, and social media integration.

Accolades

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| 2019 | 2nd Place , Goldman Sachs Marquee Datathon | New York, NY |
| 2018 | Winner , Google Travel and Conference Grant for LISA18 | Nashville, TN |
| 2017 | Finalist , OpenBracket Delaware Coding Championship | Wilmington, DE |
| 2017 | 8/15 Pts. , American Invitational Mathematics Examination | Atlanta, GA |
| 2016 | 9th Place , Harvard-MIT Math Tournament Sweepstakes | Cambridge, MA |
| 2016 | 10th Place , Harvard-MIT Math Tournament Team | Cambridge, MA |
| 2016 | 6th Place , National Science Olympiad Medalist | Menomonie, WI |

Skills

Languages Python, JAVA, C, LC3 Assembly, SQL, R
Technologies Android Studio, Git, $\text{\LaTeX}$, Linux, Unix